

Customer Churn Prediction Using Machine Learning

TCS Industry Project

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The Business Imperative: Identifying At-Risk Customers

The core objective is to predict customer churn (Yes/No) using historical data, transitioning from reactive damage control to a proactive early-warning system.



Proactive Retention

Shift strategy to intervene before the customer leaves.



Resource Optimization

Target retention budgets specifically toward high-risk profiles.



Strategic Value

Improve Customer Lifetime Value (CLV) by preventing silent attrition.

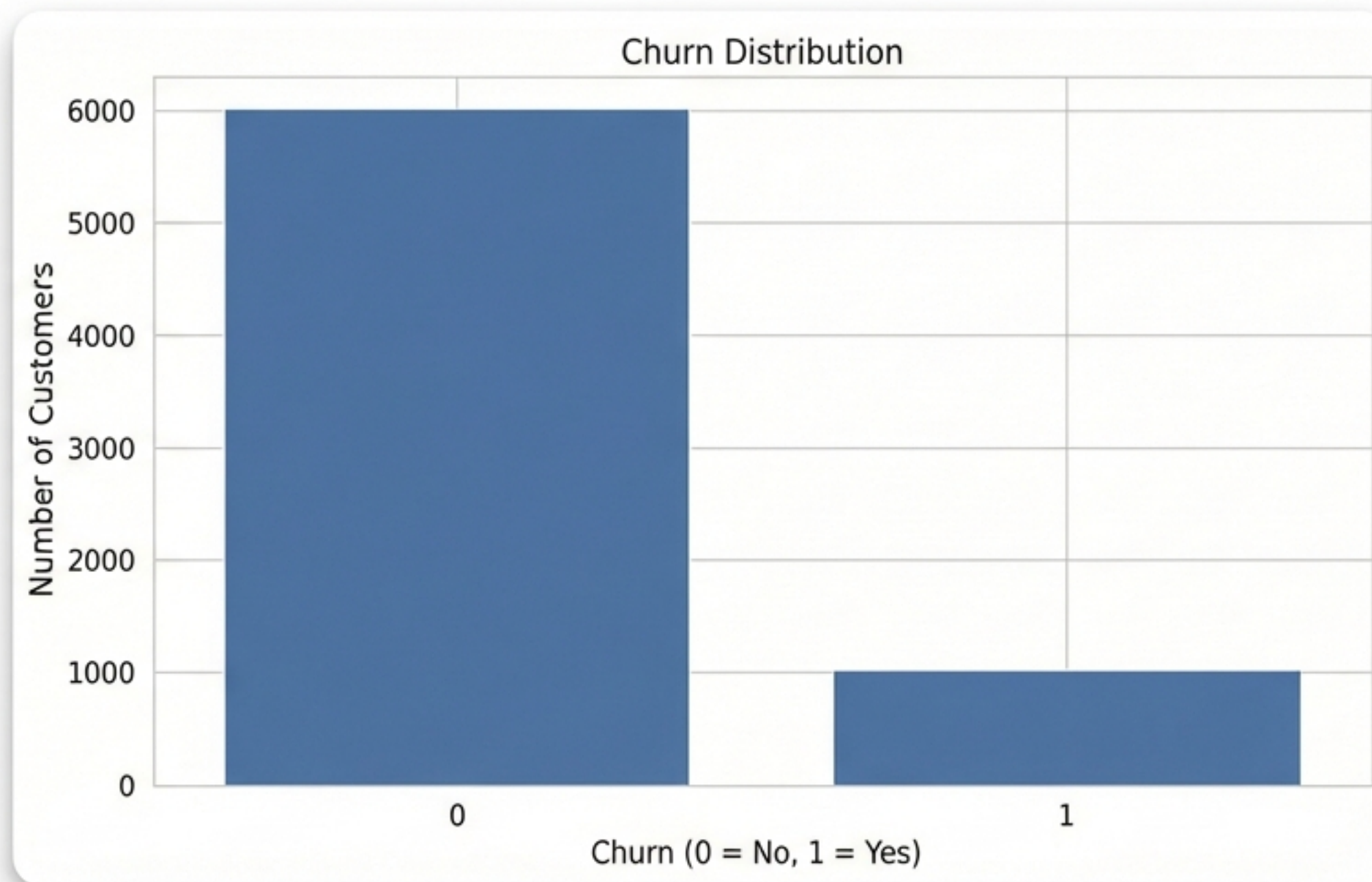
Dataset Reality: The Challenge of Class Imbalance

Data Profile

- **Volume:** 10,000 customer records
- **Complexity:** 32 distinct features (demographic, behavioral, engagement)
- **Target:** Binary classification (0 = No Churn, 1 = Yes Churn)

Key Insight

The dataset is heavily imbalanced. The vast majority of users are non-churners, creating a risk of false accuracy where models ignore the minority class.



Technical Methodology & Pipeline



Model Selection Strategy

Model A: Logistic Regression

- A linear model chosen for interpretability.
- Critical Adjustment: **`class_weight='balanced'`**
- Why: This parameter penalizes the model heavily for missing the minority class (Churn), prioritizing Recall over raw Accuracy.

VS

Model B: Random Forest Classifier

- A powerful ensemble method using decision trees.
- Configuration: Standard implementation.
- Why: Chosen as a benchmark for its ability to handle non-linear relationships and high-dimensional data.

Logistic Regression: Prioritizing Recall

Accuracy: 66.8%

Recall (Class 1): 0.66

The model successfully identifies 66% of at-risk customers (134 out of 204). While precision is lower, the high recall ensures we cast a wide net to catch potential churners.

Logistic Regression				
0.668				
[[1202 594]				
[70 134]]				
	precision	recall	f1-score	support
0	0.94	0.67	0.78	1796
1	0.18	0.66	0.29	204
accuracy			0.67	2000
macro avg	0.56	0.66	0.54	2000
weighted avg	0.87	0.67	0.73	2000

Random Forest: The Accuracy Trap

Accuracy: 89.75%

Recall (Class 1): 0.02

Despite high accuracy, the model fails to detect churners, finding only 4 out of 204 cases. It optimizes for the majority class, making it operationally useless for retention.

Random Forest					
0.8975					
[[1791 5]					
[200 4]]					
	precision	recall	f1-score	support	
0	0.90	1.00	0.95	1796	
1	0.44	0.02	0.04	204	
accuracy			0.90	2000	
macro avg	0.67	0.51	0.49	2000	
weighted avg	0.85	0.90	0.85	2000	

Verdict & Future Scope

Final Decision

Winner: Logistic Regression



Rationale: In churn prediction, a False Negative (missing a churner) is more costly than a False Positive. We prioritize Recall over Accuracy. Accuracy is a misleading metric for imbalanced datasets.

Future Scope

- **Feature Engineering:** Develop interaction features to improve precision.
- **Advanced Modeling:** Test XGBoost or Gradient Boosting with hyperparameter tuning.
- **Deployment:** Transition from notebook analysis to a real-time API prediction system.